Chaitanya Arora

New Delhi, India — +91-9599758527 — chaitanyaarora2006@gmail.com
LinkedIn — GitHub

Professional Summary

Backend AI Engineering enthusiast and B.Tech Computer Science student with experience in FastAPI, React, Node.js, Python, and AI/ML. Passionate about building scalable backend systems, REST APIs, LLM-powered applications, and deep learning solutions, with strong problem-solving and DSA skills.

Education

Amity University, Noida 2024 – 2028
Bachelor of Technology in Computer Science Engineering
CGPA: **8.72/10** (4th Semester)

Class XII (CBSE): 82.4%
Class X (CBSE): 86%

Technical Skills

Languages	Python, Java, C, C++, JavaScript
Frontend	React.js, HTML, CSS
Backend	Node.js, Express.js, REST APIs, FastAPI
Databases	MongoDB, MySQL
AI & LLM	Large Language Models (LLMs), LangChain, RAG, Prompt Engineering
Tools	Git, GitHub, Postman, VS Code, Linux
Core Concepts	Data Structures & Algorithms, Machine Learning, Deep Learning

Projects

AI-Based Video Anomaly Detection System *PyTorch, YOLOv8, React, Express.js, MongoDB, Cloudinary*
GitHub

- Designed an AI-powered video anomaly detection pipeline using a **BiGRU + Multi-Head Attention** architecture trained on the **UCF Crime** dataset for automated CCTV surveillance analysis.
- Integrated **YOLOv8** for object detection and **EfficientNet-B0** for feature extraction to generate anomaly confidence scores, timestamps, and localized event highlighting.
- Developed a full-stack web application using **React, Node.js, Express.js, MongoDB, and Cloudinary** for video uploads, cloud storage, AI inference, and interactive anomaly visualization.

FunTube *React.js, Node.js, Express.js, MongoDB*
GitHub View

- Developed a full-stack video-sharing application inspired by YouTube, enabling users to create, upload, and play videos.
- Built a responsive user interface with **React.js** and implemented backend services and REST APIs using **Node.js** and **Express.js**.
- Integrated **MongoDB** for application data management and deployed the frontend on **Vercel**.

AI-Based Melanoma Detection System *Python, TensorFlow, Keras*

- Developed a convolutional neural network (CNN) to classify skin lesions from dermoscopic images.
- Applied image preprocessing, augmentation, and normalization techniques to improve model robustness and generalization.
- Built an end-to-end deep learning pipeline for automated melanoma detection using TensorFlow and Keras.

Achievements

- Smart India Hackathon (SIH) 2025 – Internal Finalist
- Participated in Cyber Cup 5.0 (Amity University)
- Participated in AlgoVerse HackWithIndia Hackathon

Areas of Interest

Backend Development • Artificial Intelligence • Large Language Models • Machine Learning • Full Stack Development • Data Structures & Algorithms